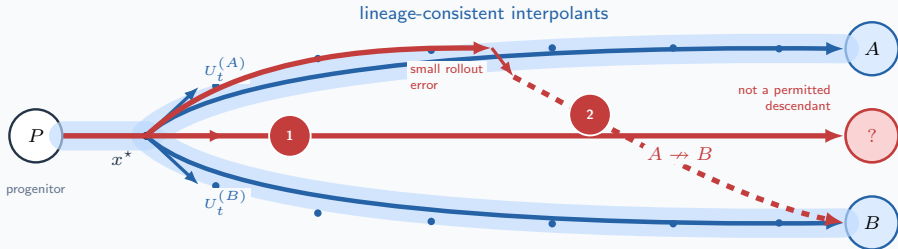


Why constrain the learned trajectories?

lineage-consistent training interpolants do not guarantee lineage-consistent generated trajectories



1. Conditional averaging

$v^* = \mathbb{E}[U_t \mid I_t = x^*]$ can point between valid lineage branches.

2. Off-support extrapolation

After a rollout error, the velocity is unconstrained by training and can make an invalid $A \rightarrow B$ jump.